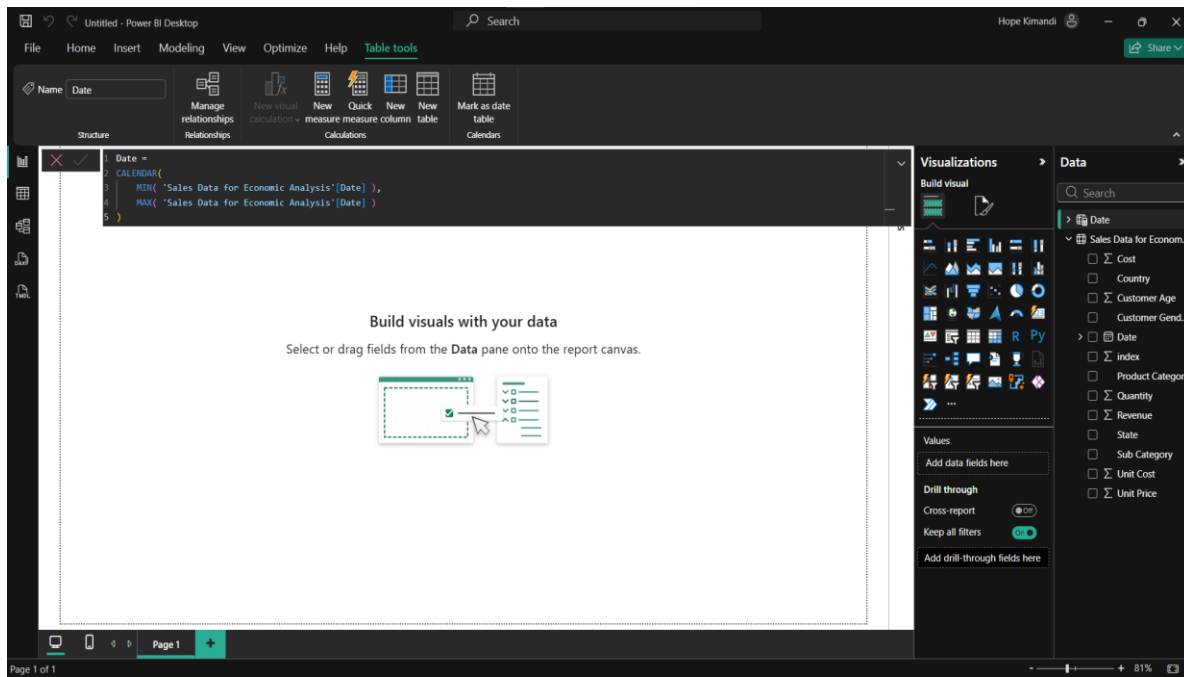
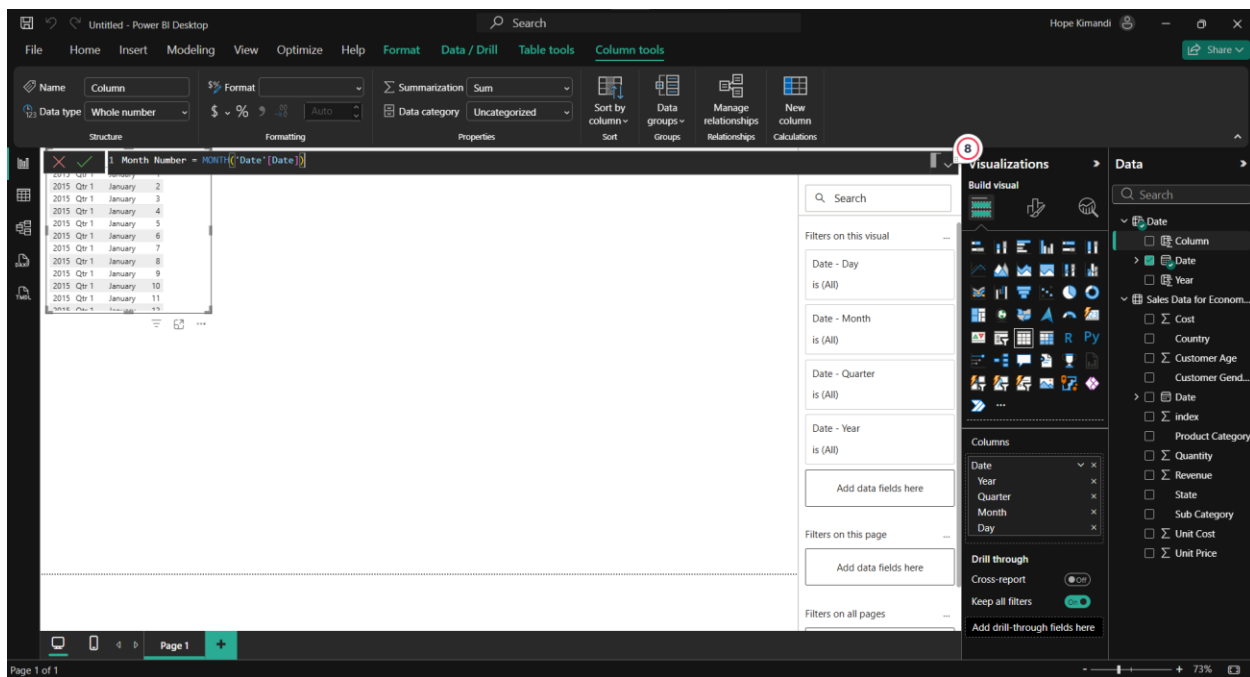


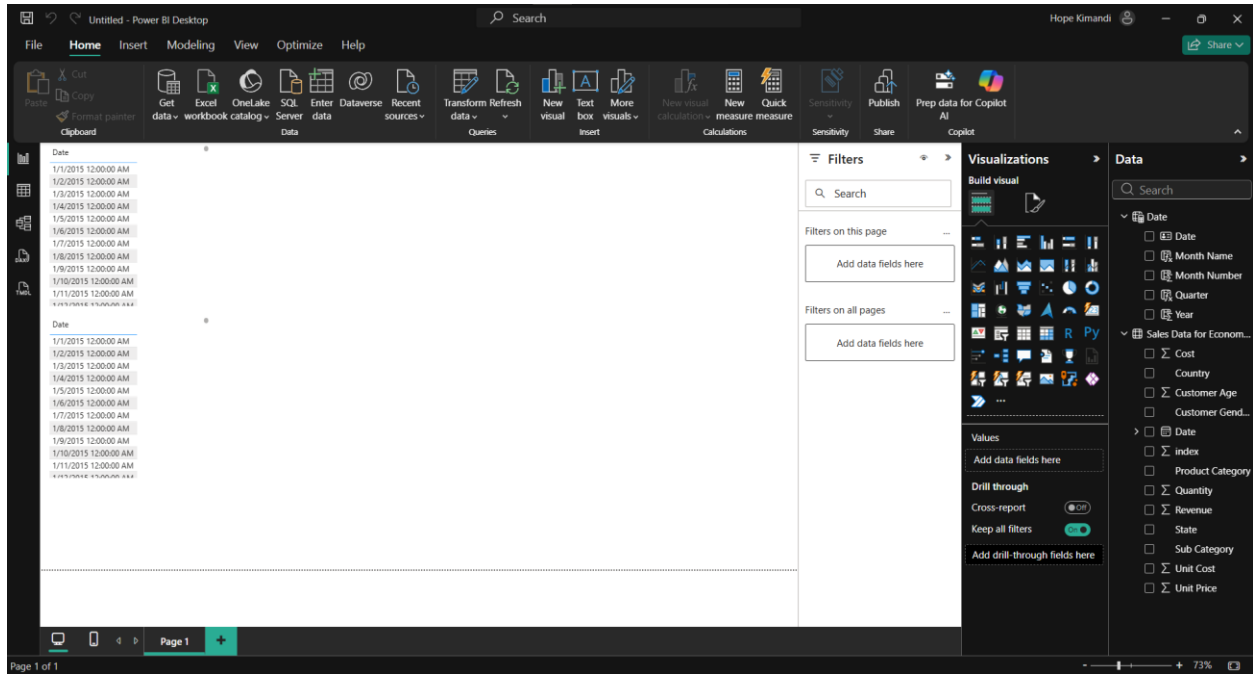
Creating a date table



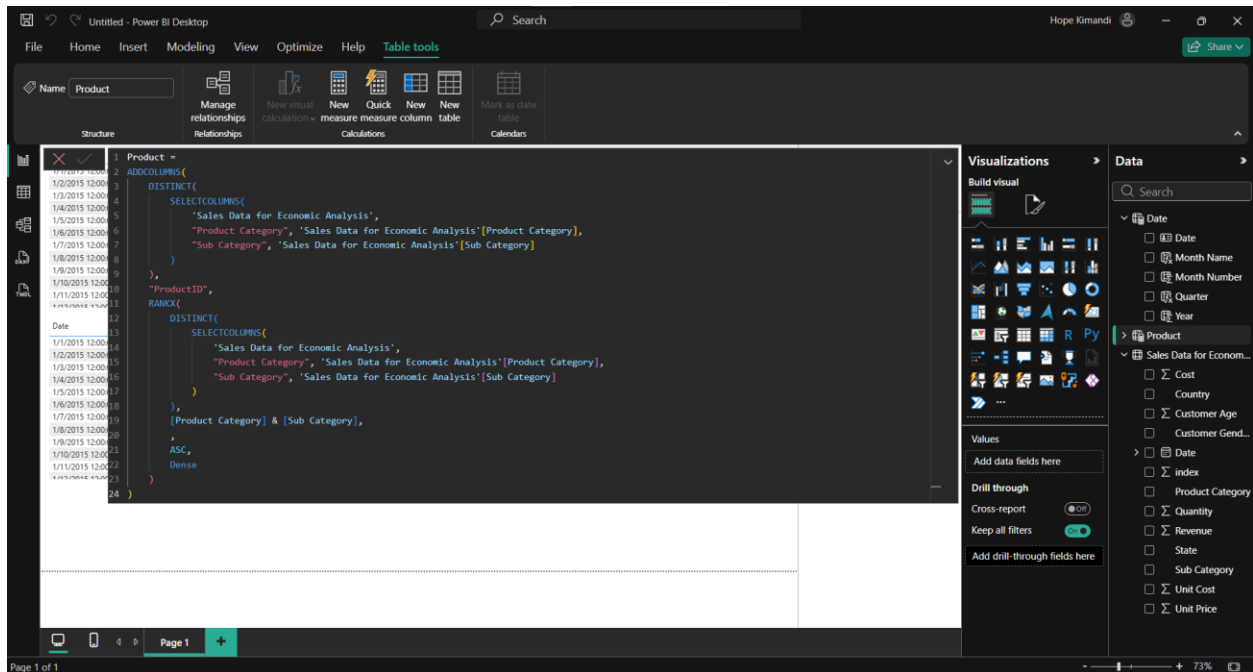
Creating columns



Final result



Creating dimensional tables



Power BI Desktop interface showing the 'Geography' table. The 'Table tools' ribbon is active, displaying options like 'Manage relationships', 'New visual calculation', 'New Quick measure', 'New column', 'New table', 'Mark as date table', and 'Calendars'. The 'Geography' table is selected in the 'Name' dropdown.

The DAX formula for the 'Geography' table is as follows:

```
Geography =
ADDCOLUMNS(
    DISTINCT(
        SELECTCOLUMNS(
            'Sales Data for Economic Analysis',
            "Country", 'Sales Data for Economic Analysis'[Country],
            "State", 'Sales Data for Economic Analysis'[State]
        )
    ),
    "GeographyID",
    RANKX(
        DISTINCT(
            SELECTCOLUMNS(
                'Sales Data for Economic Analysis',
                "Country", 'Sales Data for Economic Analysis'[Country],
                "State", 'Sales Data for Economic Analysis'[State]
            )
        ),
        [Country] & [State],
        ASC,
        Dense
    )
)
```

The 'Visualizations' pane on the right shows the 'Build visual' section with various chart types. The 'Data' pane on the right shows the 'Data' table with columns like Date, Month Name, Month Number, Quarter, Year, Country, Customer Age, Customer Gender, Cost, Revenue, State, Sub Category, Unit Cost, and Unit Price.

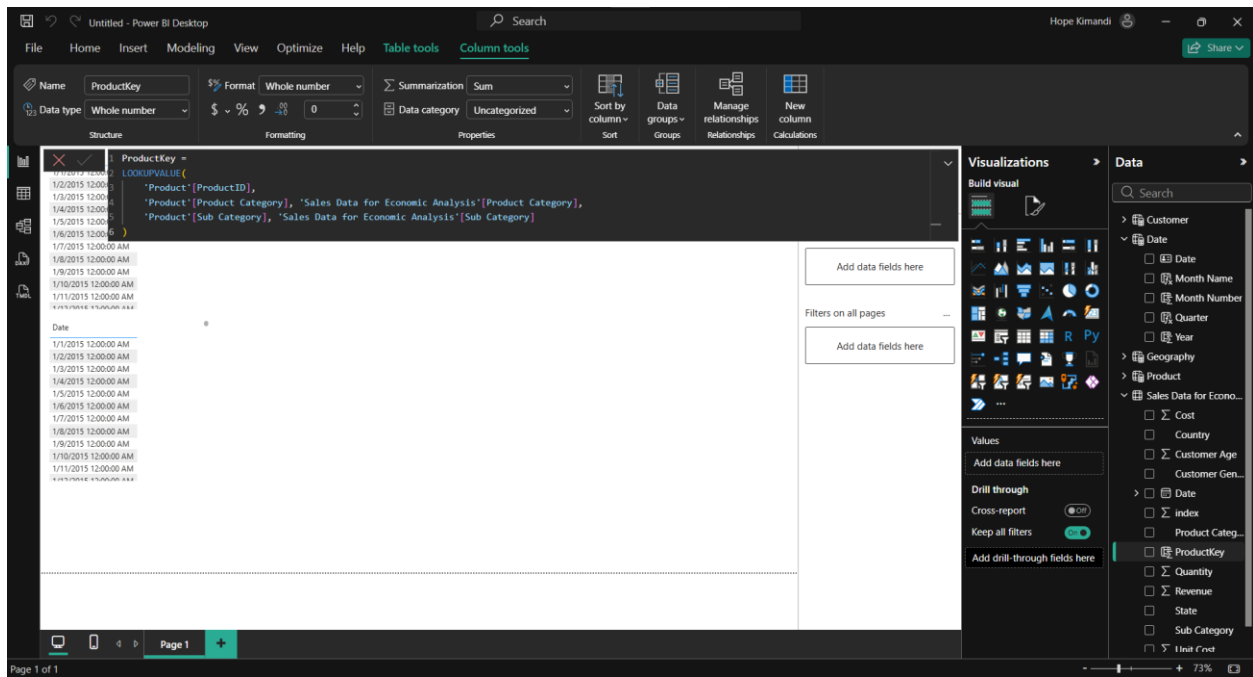
Power BI Desktop interface showing the 'Customer' table. The 'Table tools' ribbon is active, displaying options like 'Manage relationships', 'New visual calculation', 'New Quick measure', 'New column', 'New table', 'Mark as date table', and 'Calendars'. The 'Customer' table is selected in the 'Name' dropdown.

The DAX formula for the 'Customer' table is as follows:

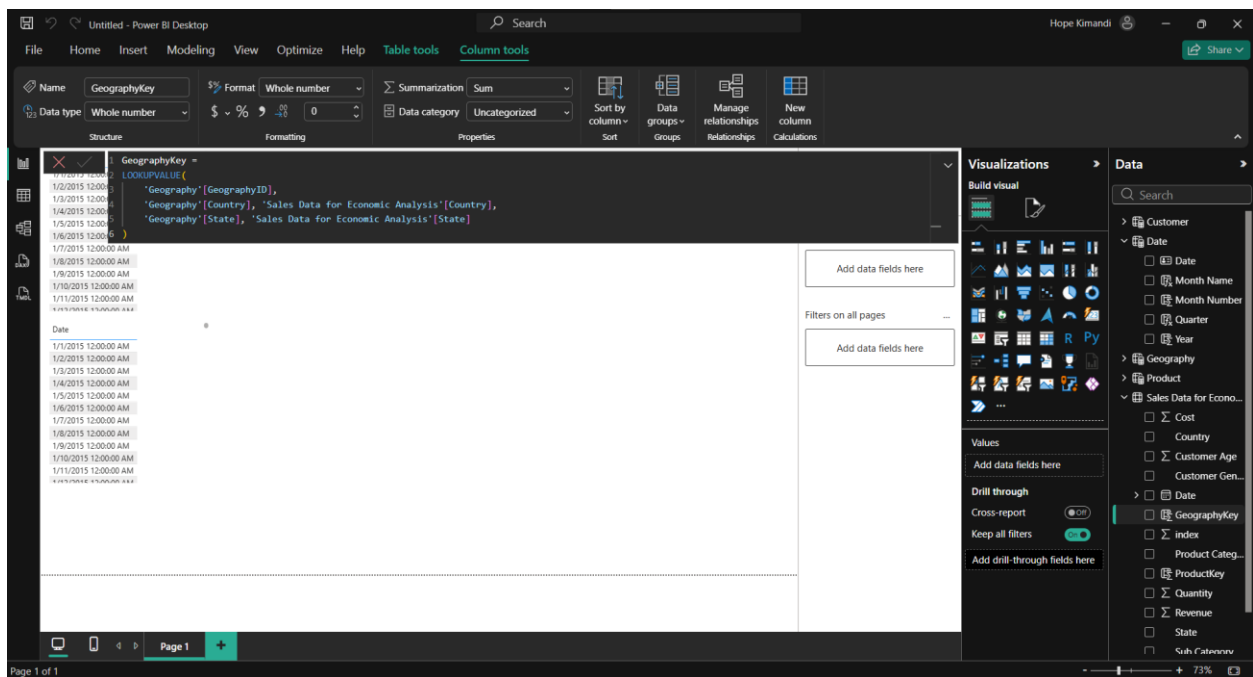
```
Customer =
ADDCOLUMNS(
    DISTINCT(
        SELECTCOLUMNS(
            'Sales Data for Economic Analysis',
            "Customer Age", 'Sales Data for Economic Analysis'[Customer Age],
            "Customer Gender", 'Sales Data for Economic Analysis'[Customer Gender]
        )
    ),
    "CustomerID",
    RANKX(
        DISTINCT(
            SELECTCOLUMNS(
                'Sales Data for Economic Analysis',
                "Customer Age", 'Sales Data for Economic Analysis'[Customer Age],
                "Customer Gender", 'Sales Data for Economic Analysis'[Customer Gender]
            )
        ),
        [Customer Age] & [Customer Gender],
        ASC,
        Dense
    )
)
```

The 'Visualizations' pane on the right shows the 'Build visual' section with various chart types. The 'Data' pane on the right shows the 'Data' table with columns like Date, Month Name, Month Number, Quarter, Year, Country, Customer Age, Customer Gender, Cost, Revenue, State, Sub Category, Unit Cost, and Unit Price.

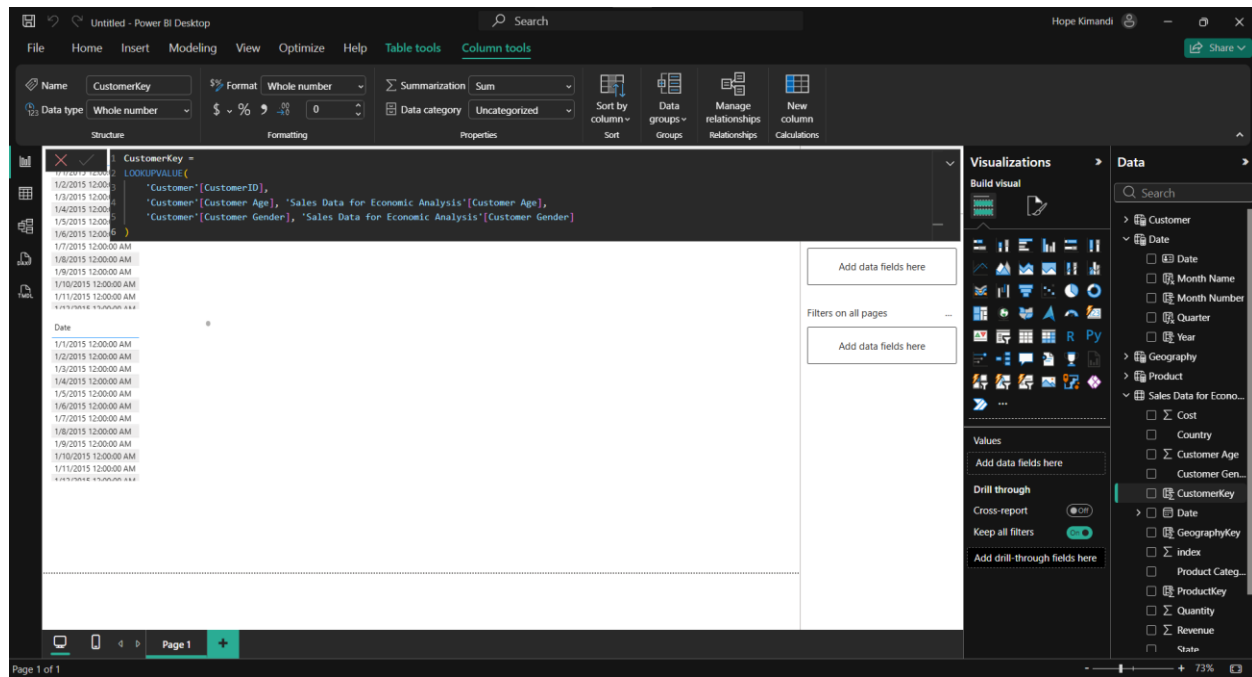
Adding product key



Add geography key

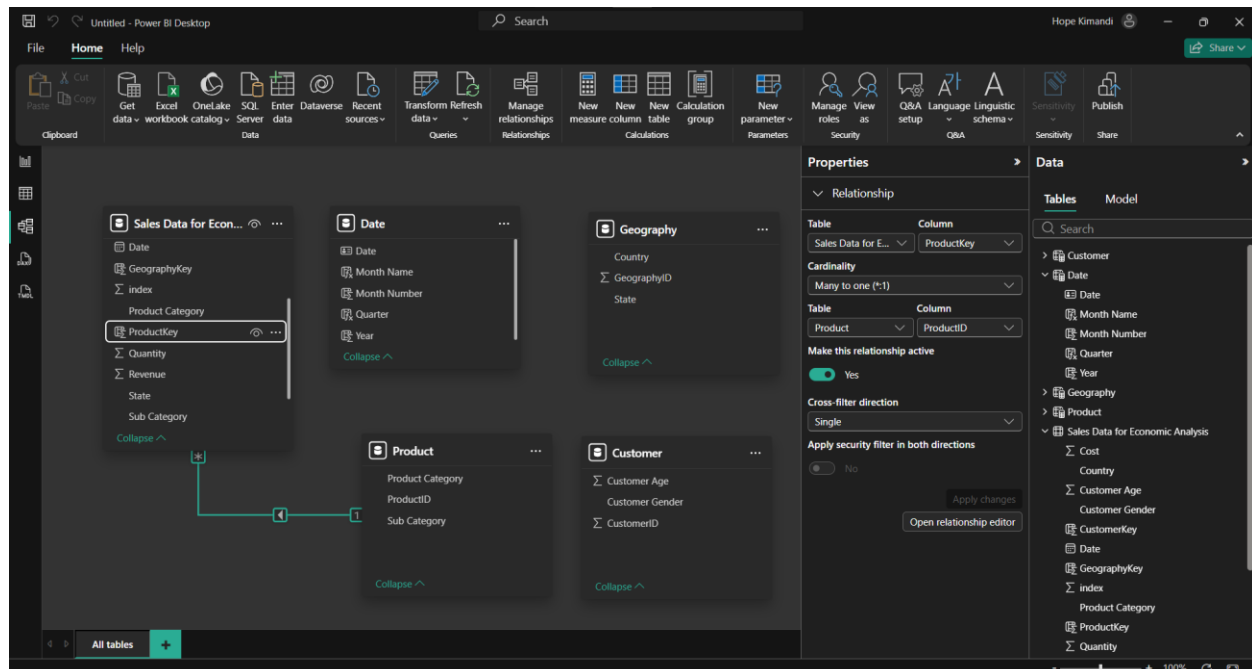


Customerkey

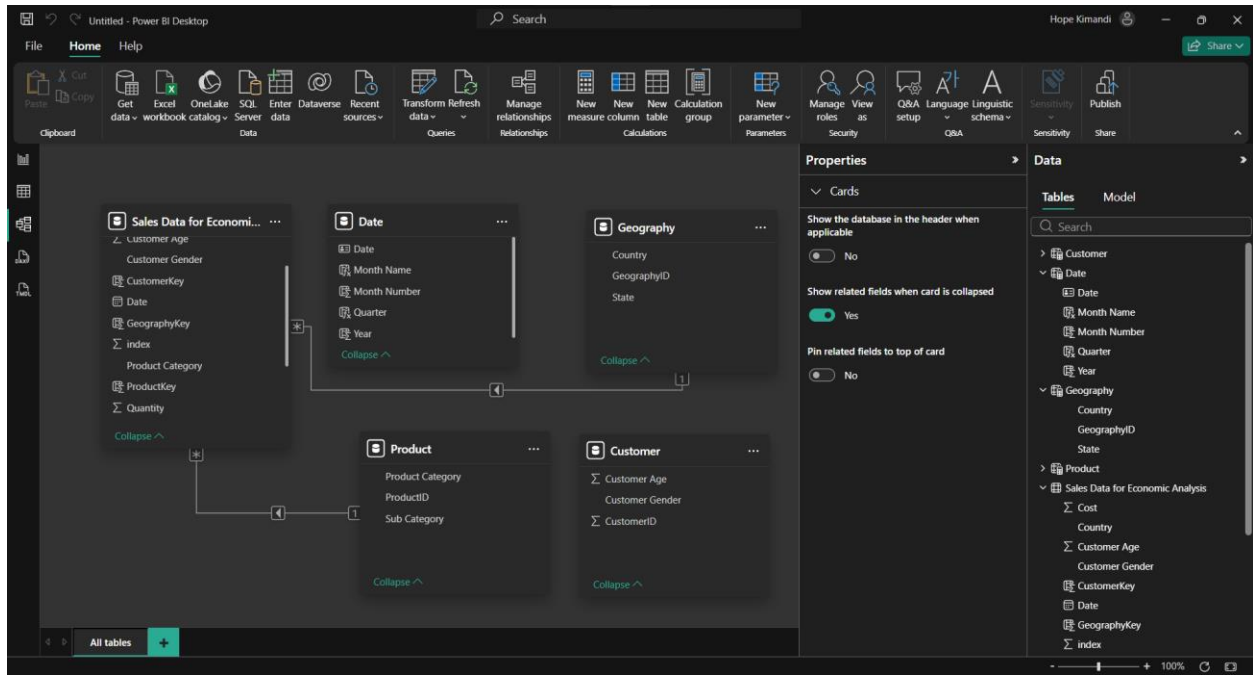


Creating relationships

a) Product and Sales Data for Economic Analysis



b) Sales Data for Economic Analysis[GeographyKey] to Geography[GeographyID]



c) Economic Analysis[CustomerKey] to Customer[CustomerID]

New relationship



Select tables and columns that are related.

From table

Sales Data for Economic Analysis

Cost	Country	Customer Age	Customer Ge...	CustomerKey	Date	Geograph
28	United States	45	M	58	Thursday, Feb...	26
87	United States	45	M	58	Wednesday, ...	26
135	United States	45	M	58	Wednesday, ...	26

To table

Customer

Customer Age	Customer Ge...	CustomerID
29	F	25
18	F	3
19	F	5

Cardinality

Many to one (*:1)

Cross-filter direction

Single



Make this relationship active



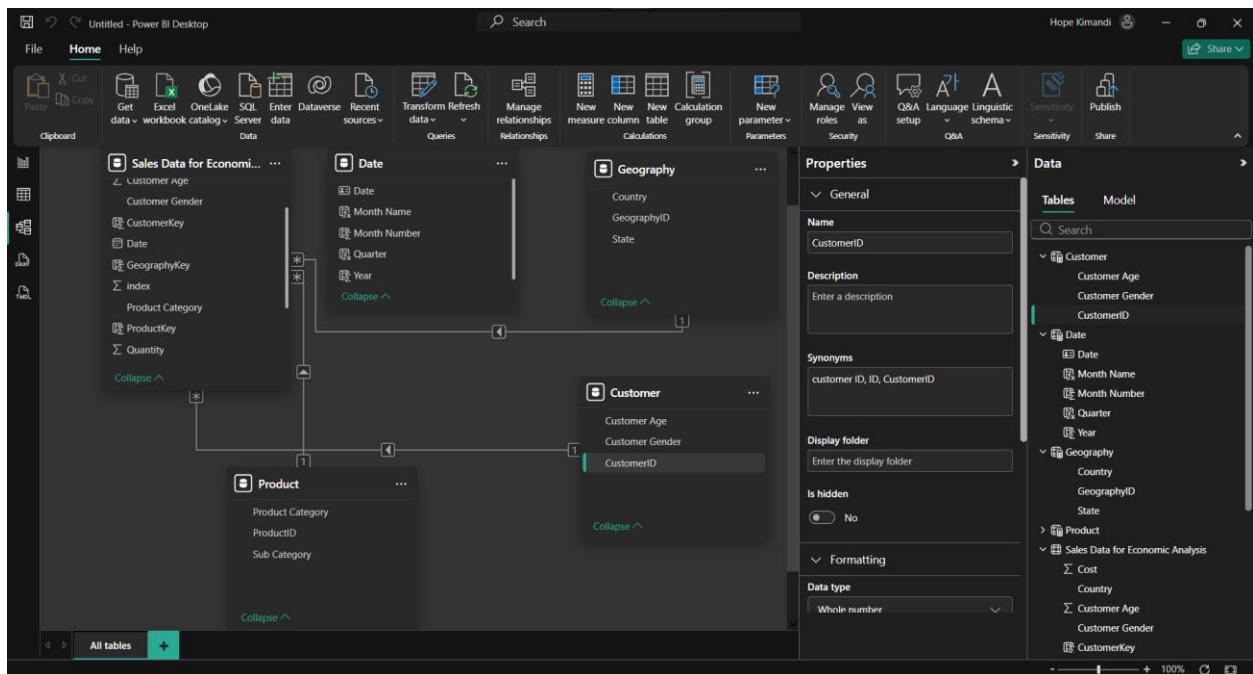
Apply security filter in both directions



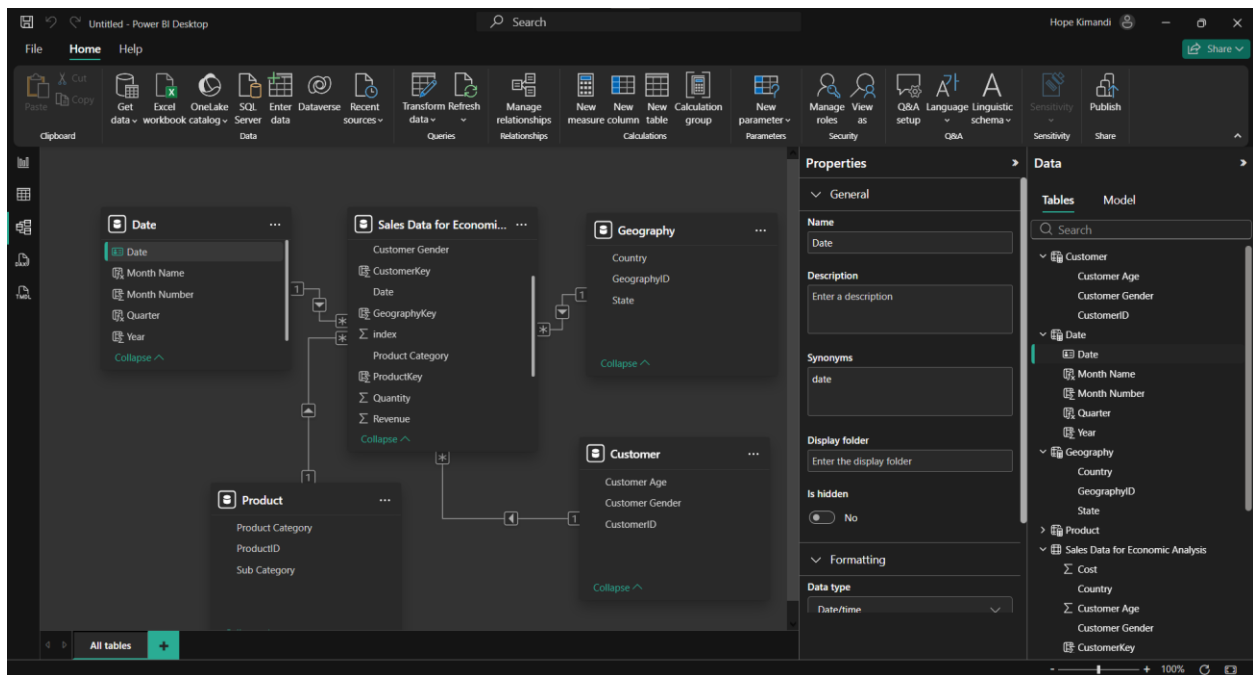
Assume referential integrity

Save

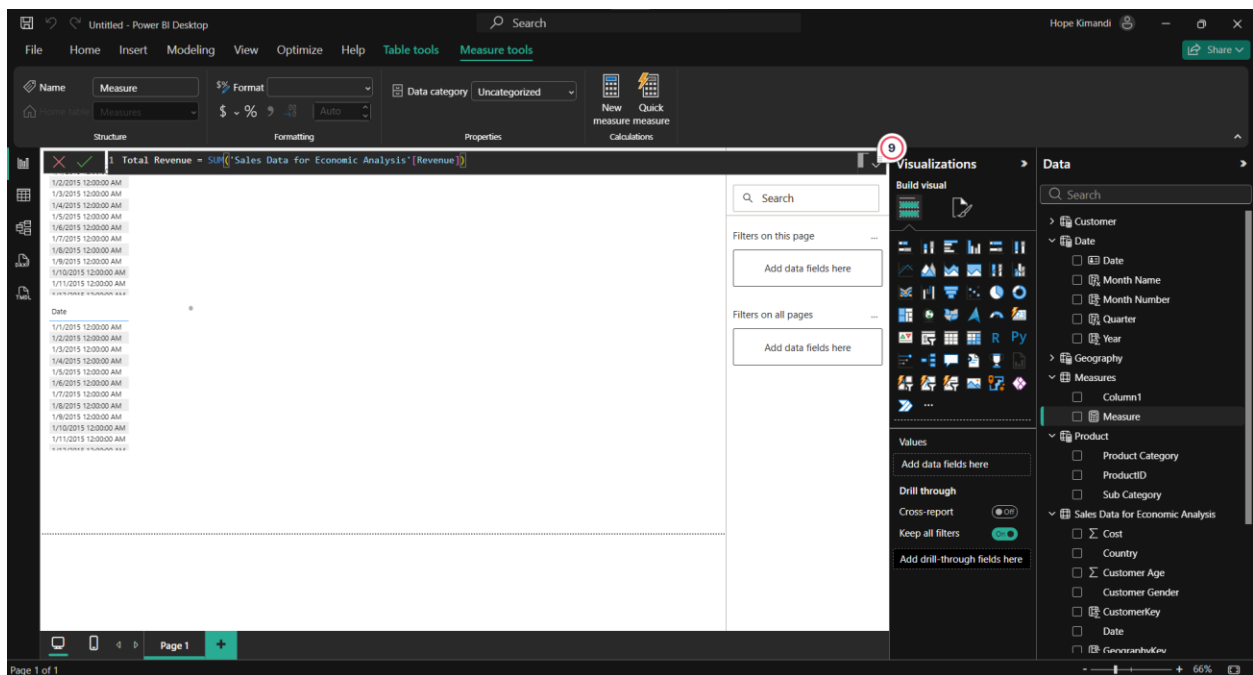
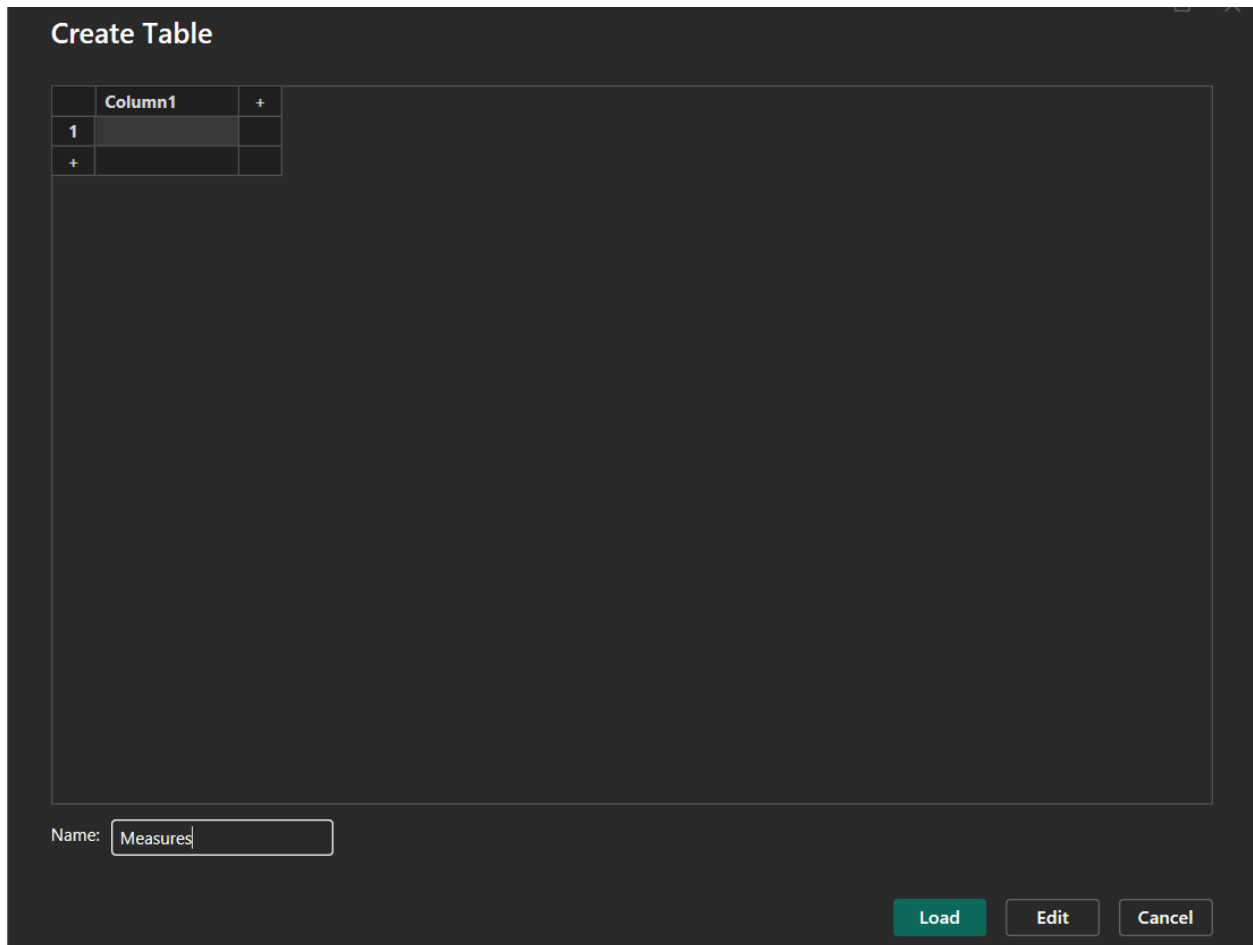
Cancel



d) Sales Data for Economic Analysis[Date] to Date[Date]



Measures



Untitled - Power BI Desktop

File Home Insert Modeling View Optimize Help Table tools Measure tools

Name: Total Cost Format: Whole number Data category: Uncategorized

Home table: Measures

Structure: Formatting Properties Calculations

Visualizations

Data

Page 1 of 1

Visualizations

Build visual

Filters on this page

Add data fields here

Filters on all pages

Add data fields here

Values

Add data fields here

Drill through

Cross-report

Keep all filters

Add drill-through fields here

Data

Customer

Date

Month Name

Month Number

Quarter

Year

Geography

Measures

Column1

Total Cost

Total Revenue

Product

Product Category

ProductID

Sub Category

Sales Data for Economic Analysis

Cost

Country

Customer Age

Customer Gender

CustomerKey

Date

Untitled - Power BI Desktop

File Home Insert Modeling View Optimize Help Table tools Measure tools

Name: Net Profit Format: Whole number Data category: Uncategorized

Home table: Measures

Structure: Formatting Properties Calculations

Visualizations

Data

Page 1 of 1

Visualizations

Build visual

Filters on this page

Add data fields here

Filters on all pages

Add data fields here

Values

Add data fields here

Drill through

Cross-report

Keep all filters

Add drill-through fields here

Data

Customer

Date

Month Name

Month Number

Quarter

Year

Geography

Measures

Column1

Net Profit

Total Cost

Total Revenue

Product

Product Category

ProductID

Sub Category

Sales Data for Economic Analysis

Cost

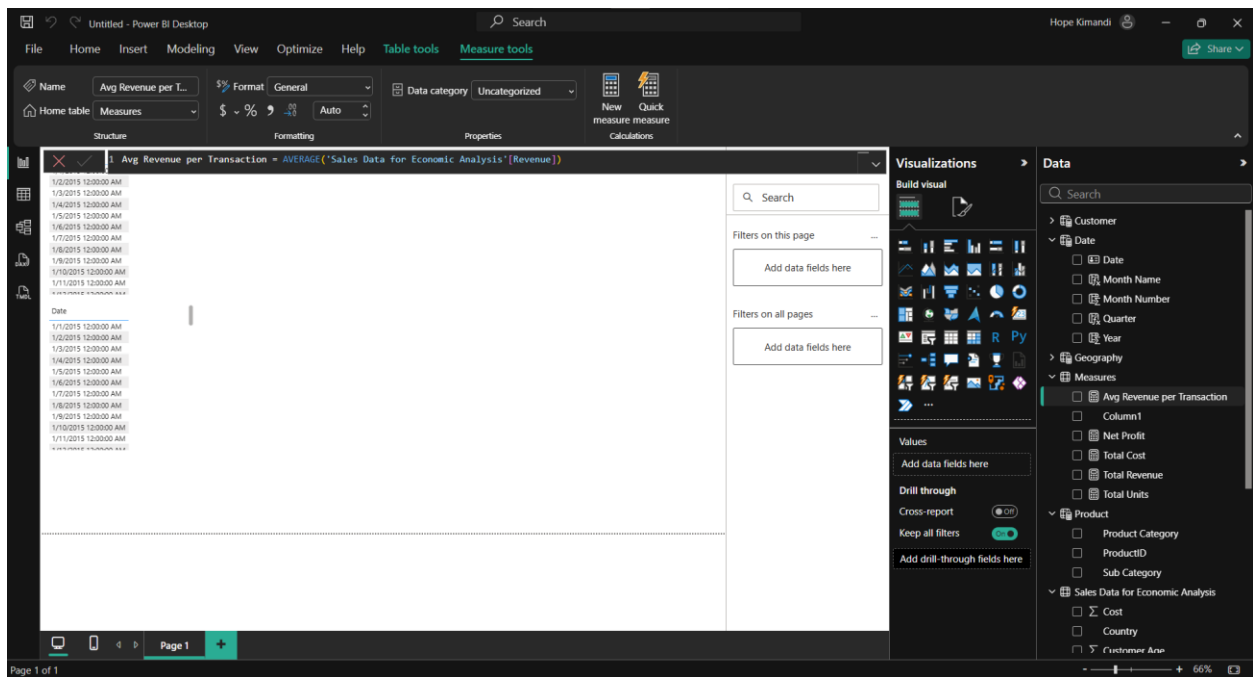
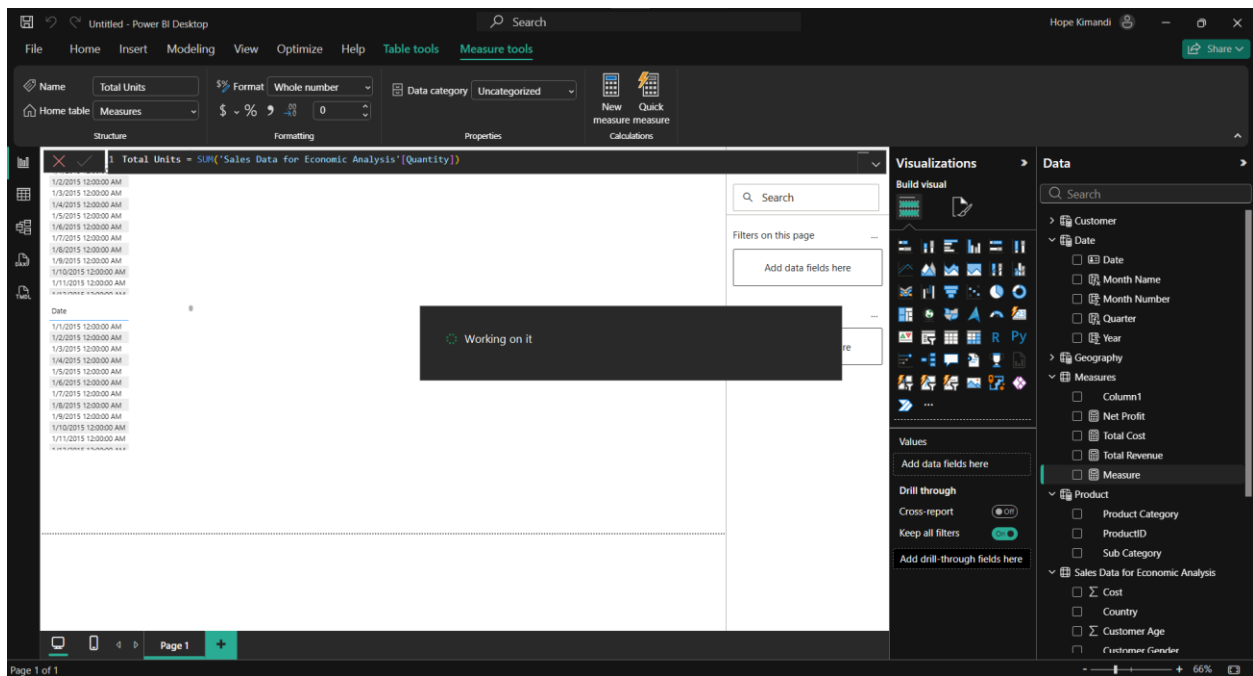
Country

Customer Age

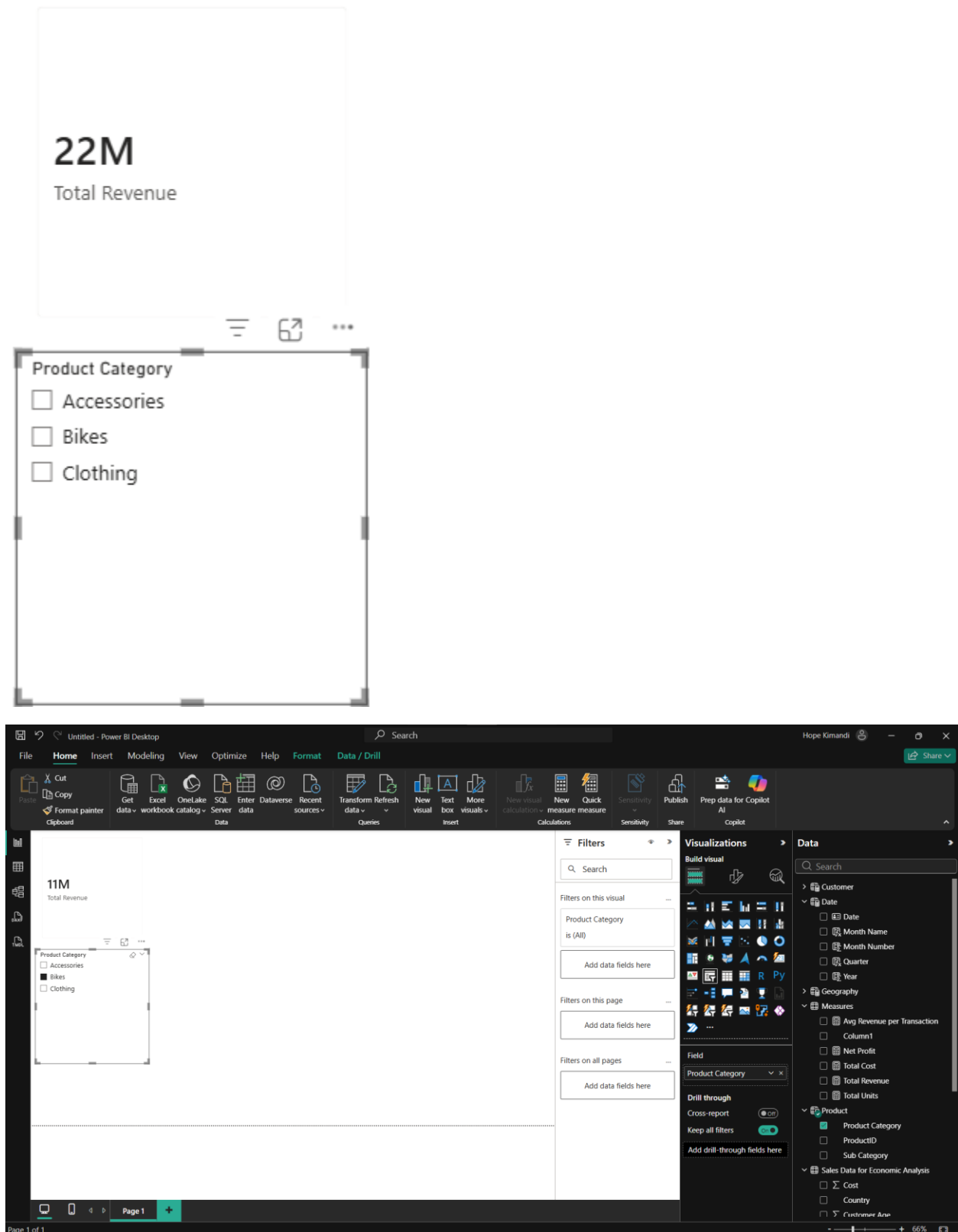
Customer Gender

CustomerKey

Date



Verify with visuals (the first page in report view)



Profit band column

The screenshot shows the Power BI Desktop interface with the 'Column tools' ribbon active. The 'Name' field is set to 'Column', 'Data type' is 'Whole number', and 'Format' is '\$ %'. The 'Summarization' dropdown is set to 'Sum', and 'Data category' is 'Uncategorized'. The 'Visualizations' pane on the right shows the 'Build visual' section with a red circle around the 'Column' icon. The 'Data' pane on the right shows the 'Sales Data for Economic Analysis' table with columns like 'Column', 'Country', 'Customer Age', etc. The main canvas displays a table with the following DAX formula for 'Profit Band':

```
Profit Band =  
VAR Margin = DIVIDE( 'Sales Data for Economic Analysis'[Revenue] - 'Sales Data for Economic Analysis'[Cost], 'Sales Data for Economic Analysis'[Revenue] )  
RETURN  
SWITCH(  
    TRUE(),  
    Margin >= 0.3, "High",  
    Margin >= 0.1, "Medium",  
    "Low"
```

The table also shows 'Total Revenue' as 22M and 'Product Category' with options for Accessories, Bikes, and Clothing.

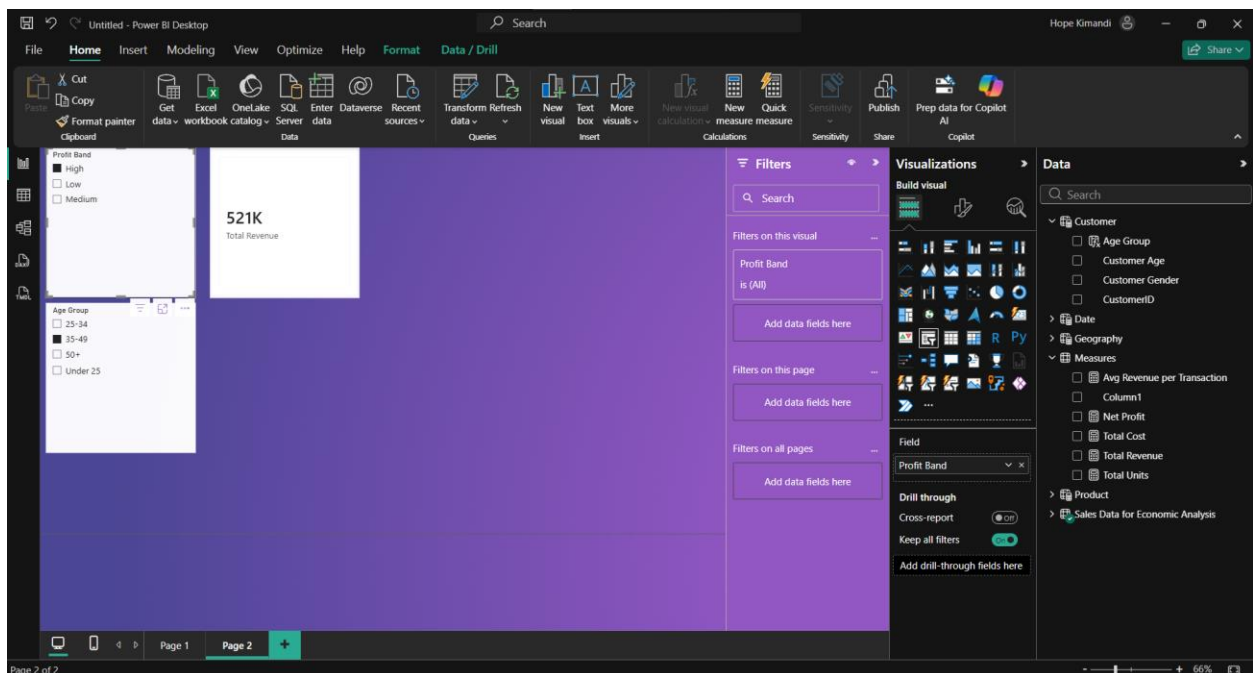
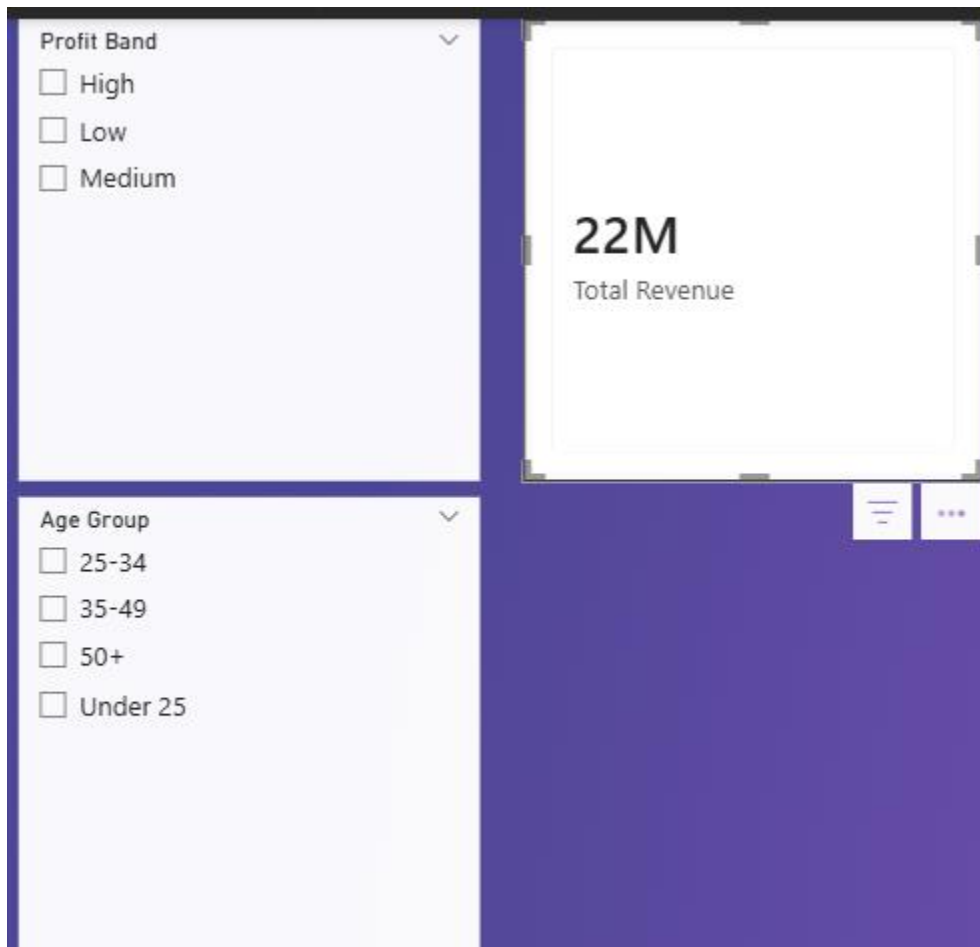
Age group column

The screenshot shows the Power BI Desktop interface with the 'Column tools' ribbon active. The 'Name' field is set to 'Age Group', 'Data type' is 'Text', and 'Format' is 'Text'. The 'Summarization' dropdown is set to 'Don't summarize', and 'Data category' is 'Uncategorized'. The 'Visualizations' pane on the right shows the 'Build visual' section with a red circle around the 'Column' icon. The 'Data' pane on the right shows the 'Sales Data for Economic Analysis' table with columns like 'Age Group', 'Customer Age', 'Customer Gender', etc. The main canvas displays a table with the following DAX formula for 'Age Group':

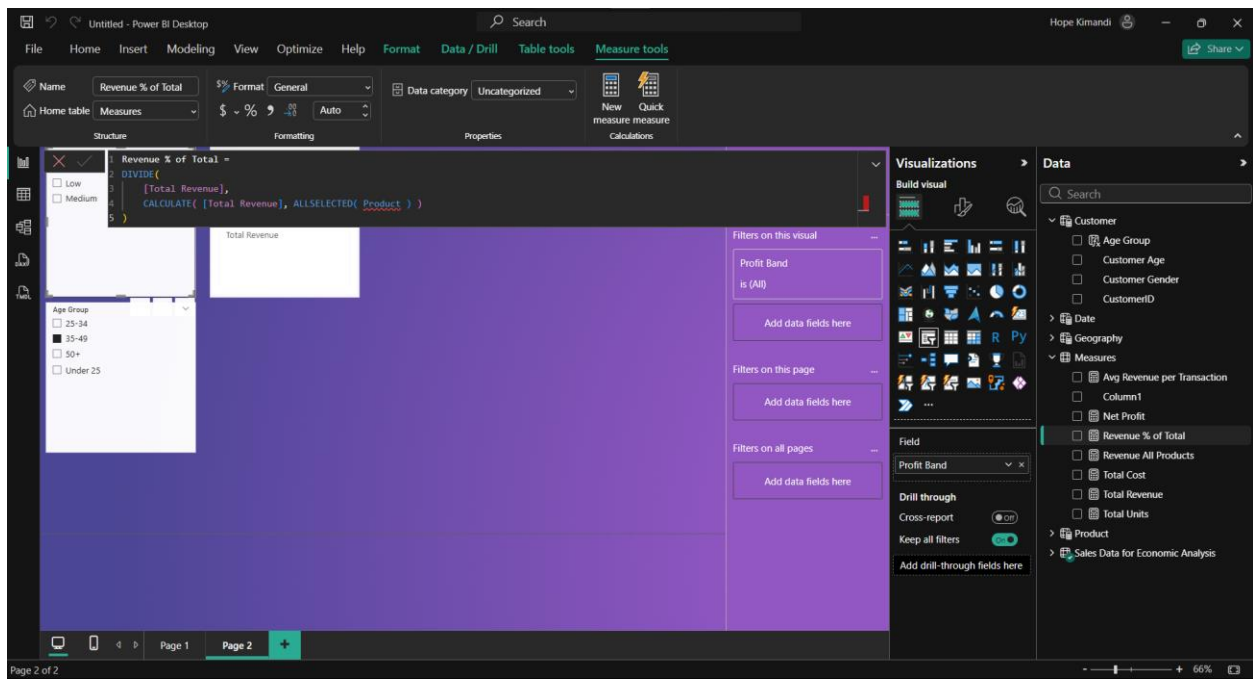
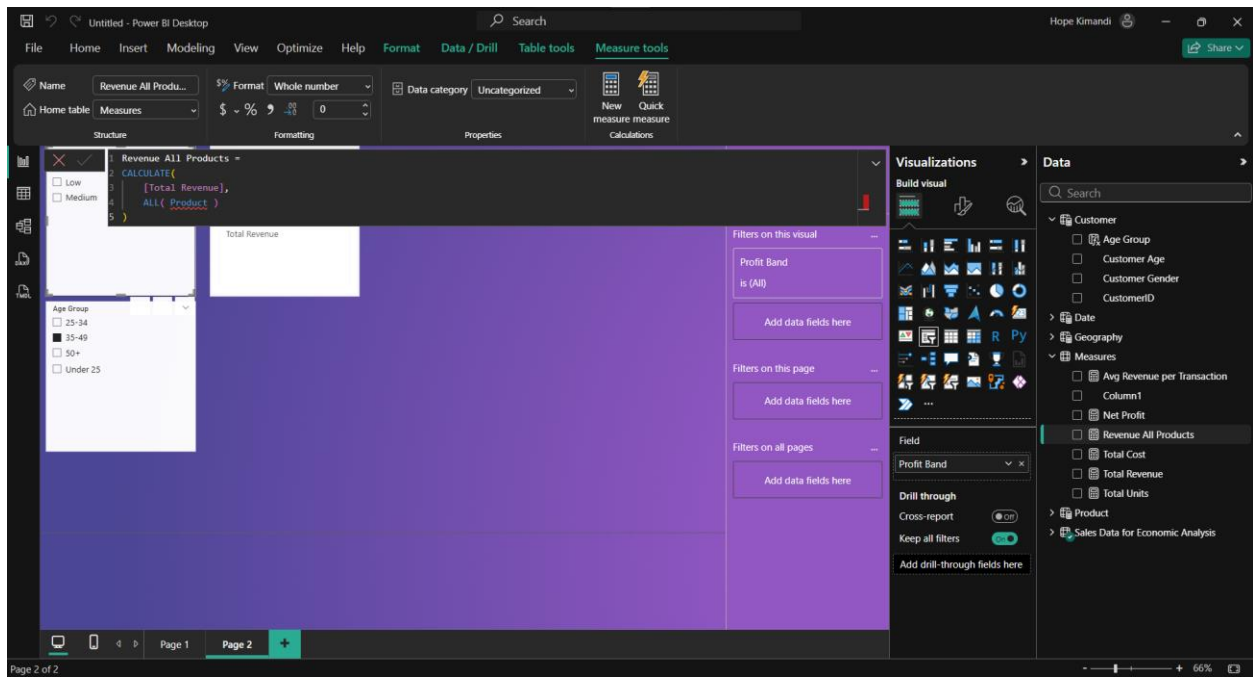
```
Age Group =  
SWITCH(  
    TRUE(),  
    'Customer'[Customer Age] < 25, "Under 25",  
    'Customer'[Customer Age] < 35, "25-34",  
    'Customer'[Customer Age] < 50, "35-49",  
    "50+"
```

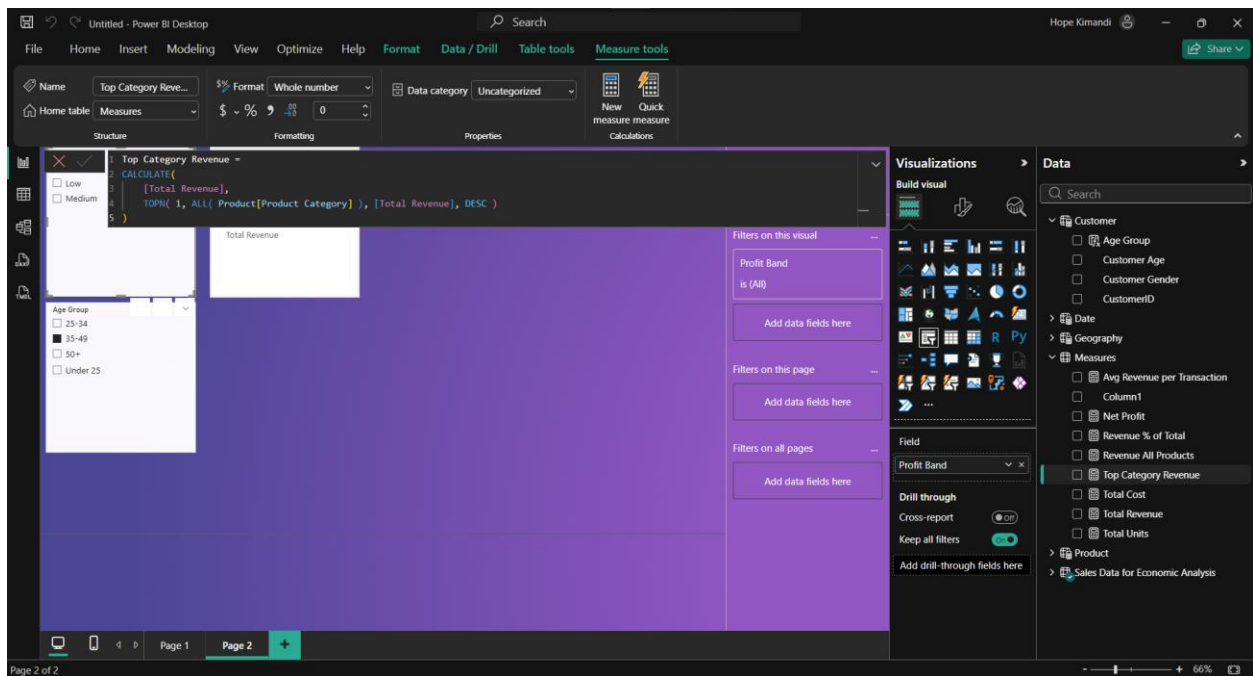
The table also shows 'Total Revenue' as 22M and 'Product Category' with options for Accessories, Bikes, and Clothing.

Verify with visuals (page 2)

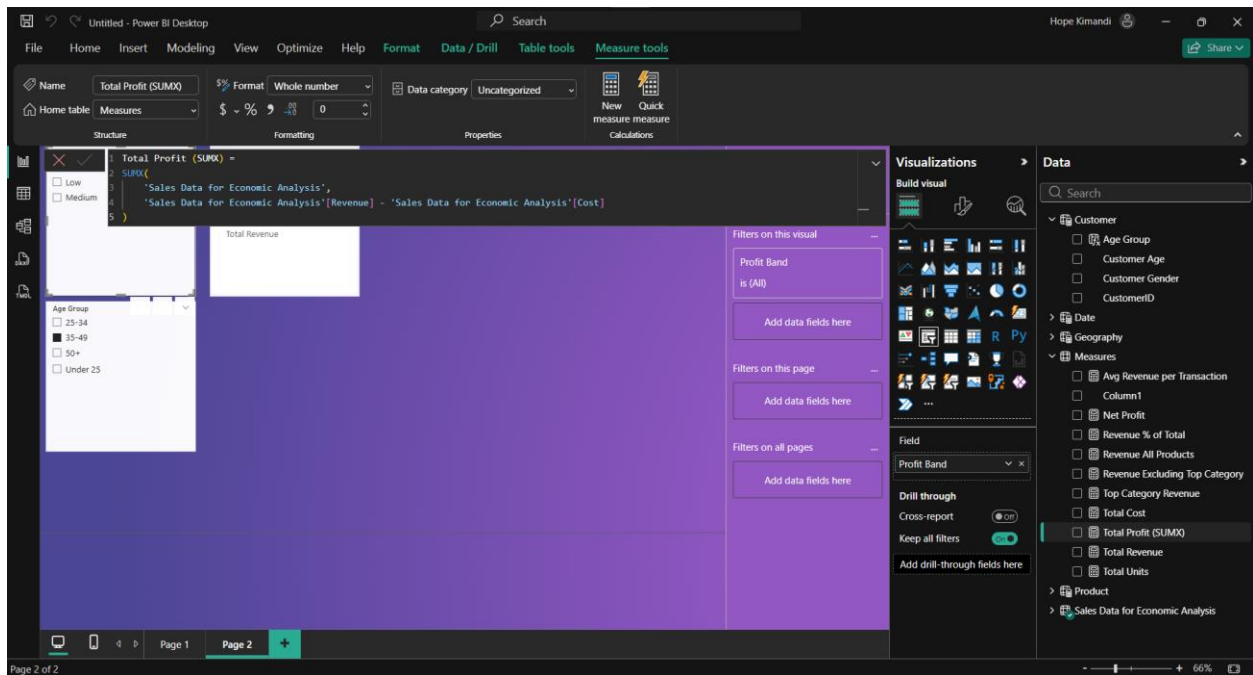


Section D





Section E



Verify with visual (page 3)

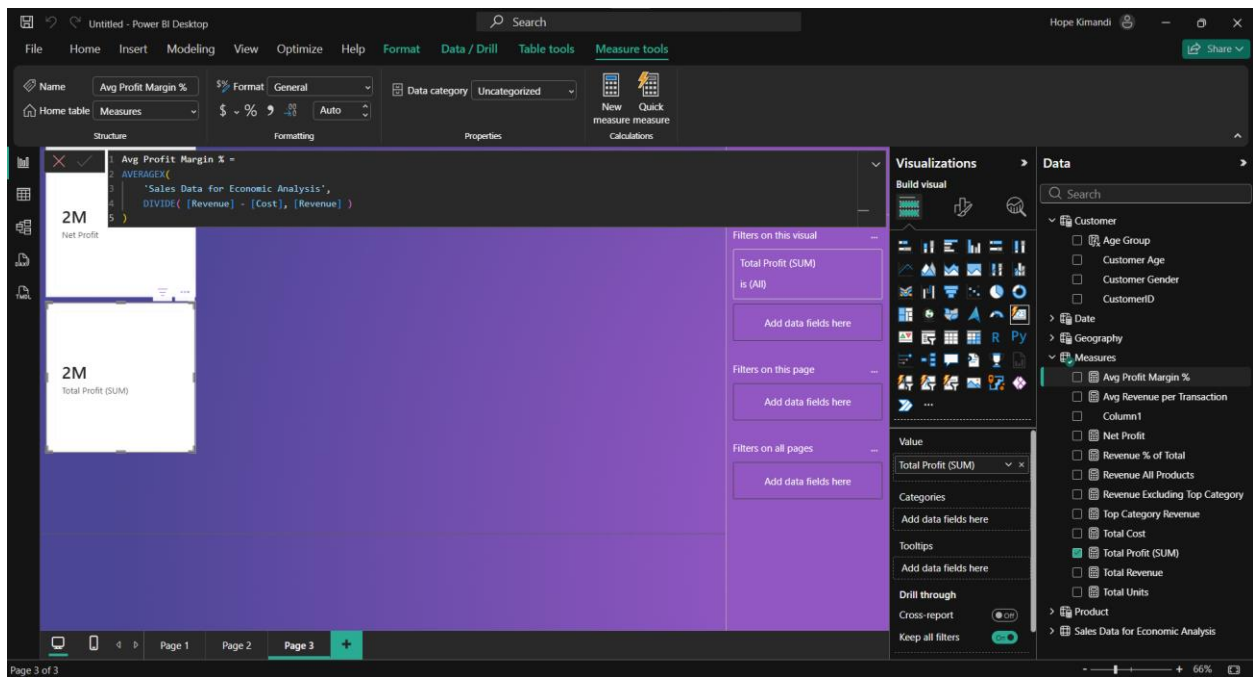
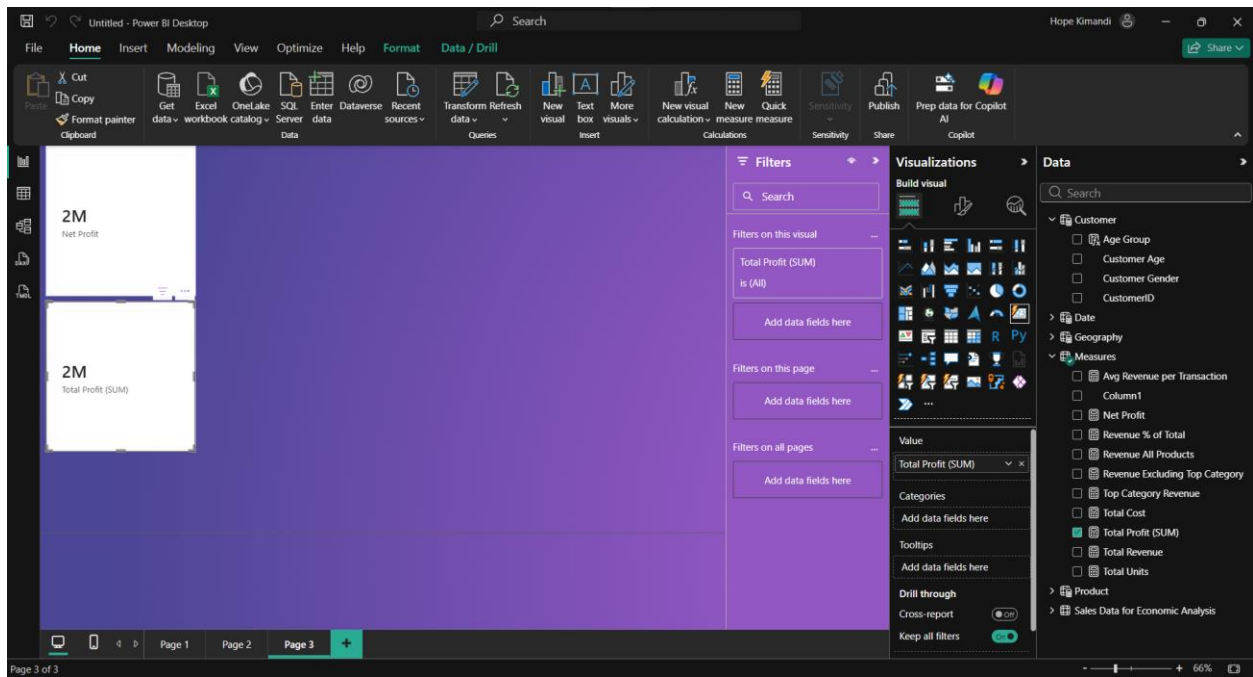
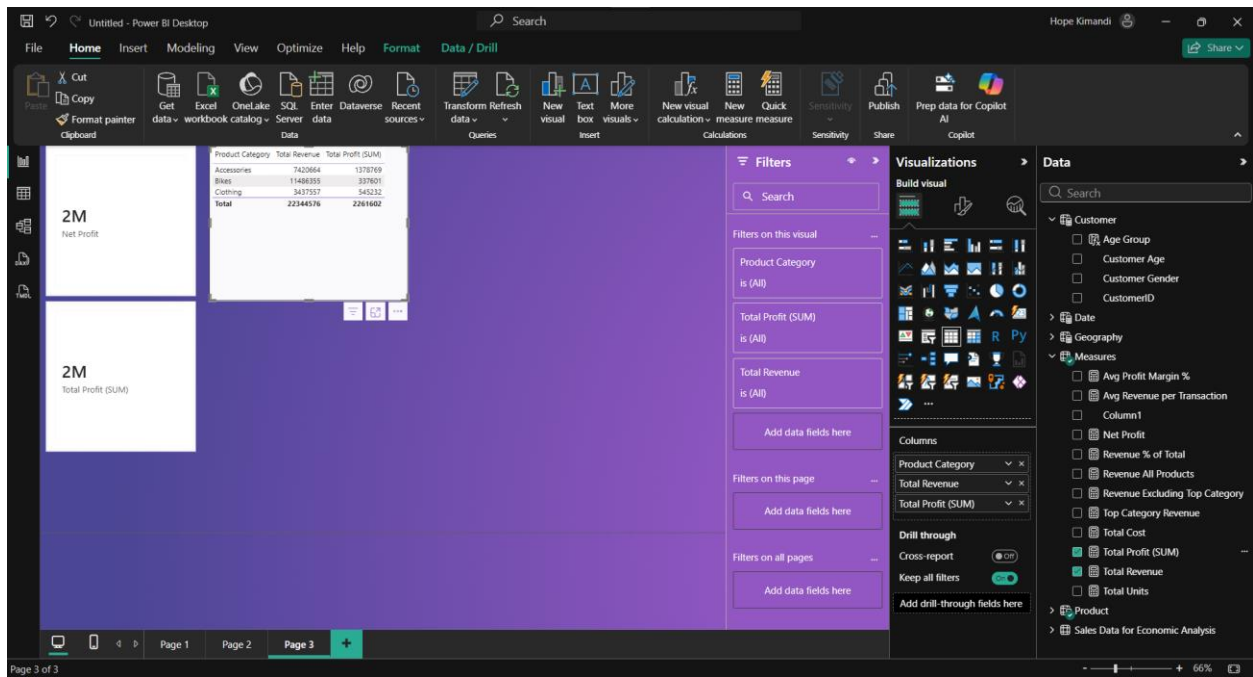
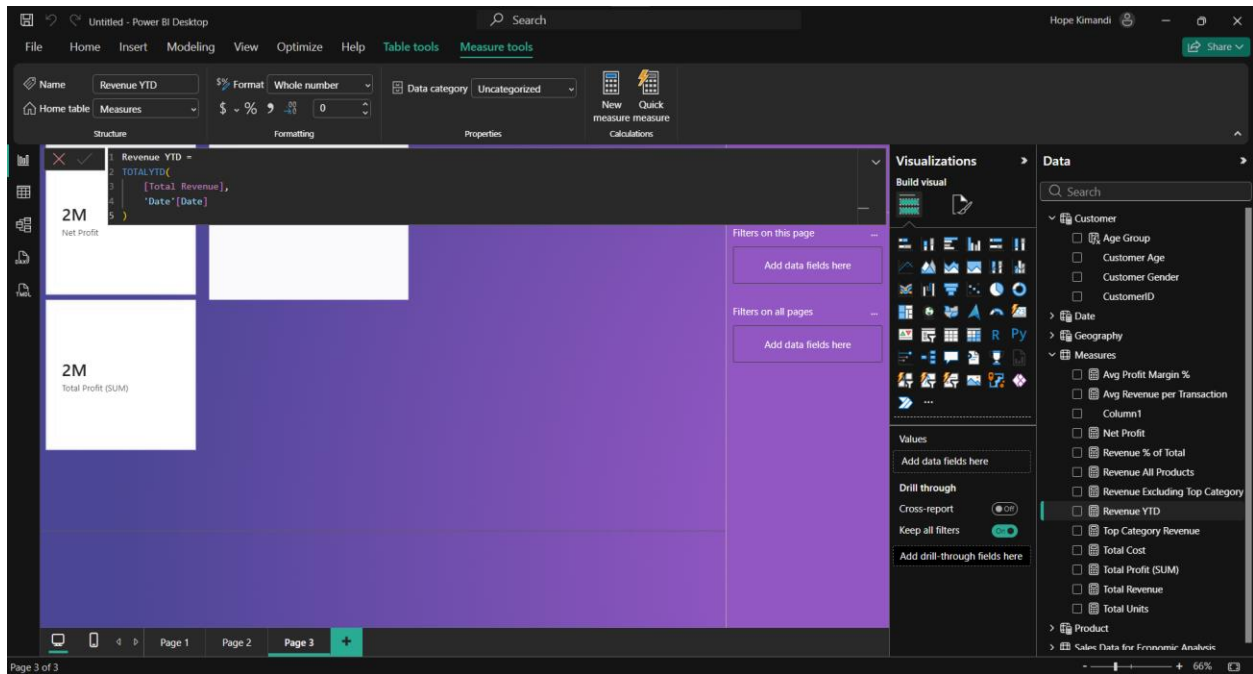
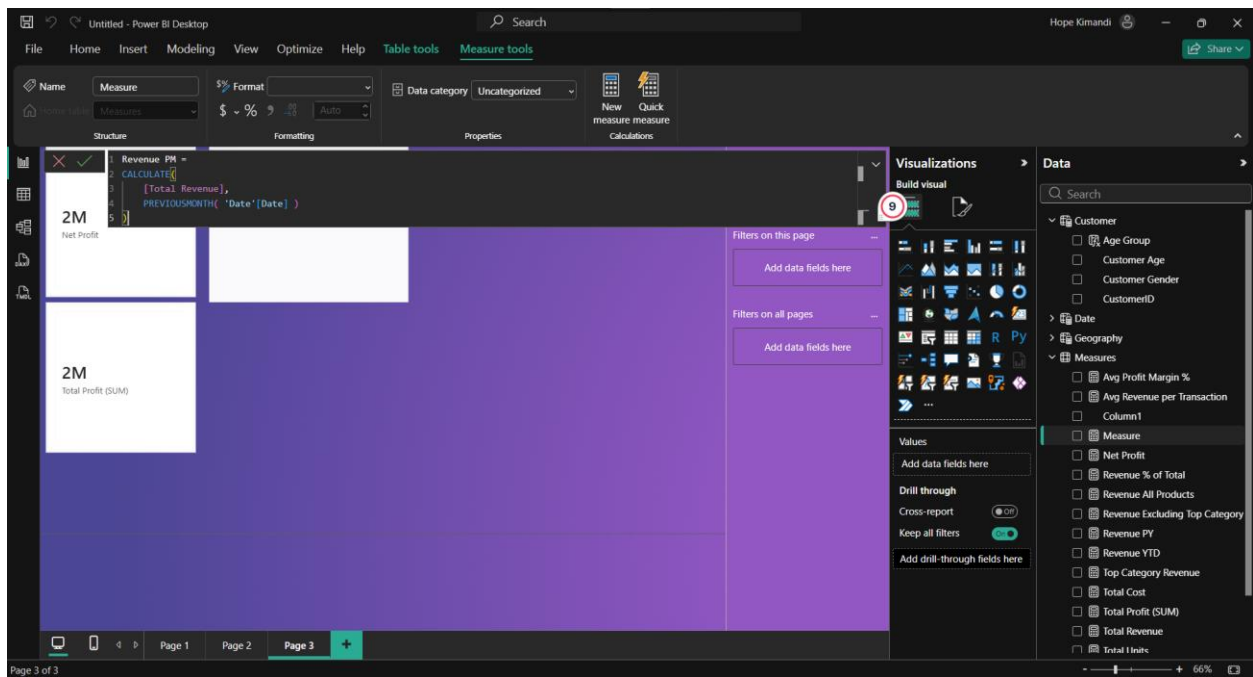
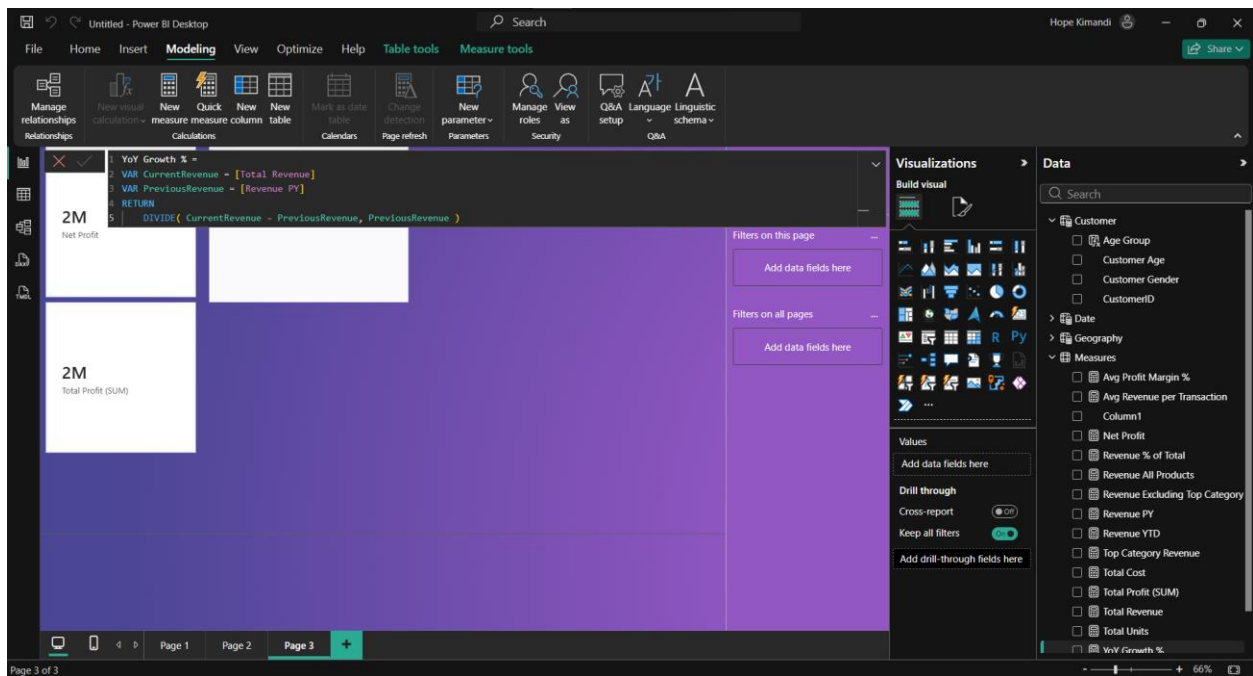


Table visual (page 3)



Section f





VISUALS IN OTHER PAGES

